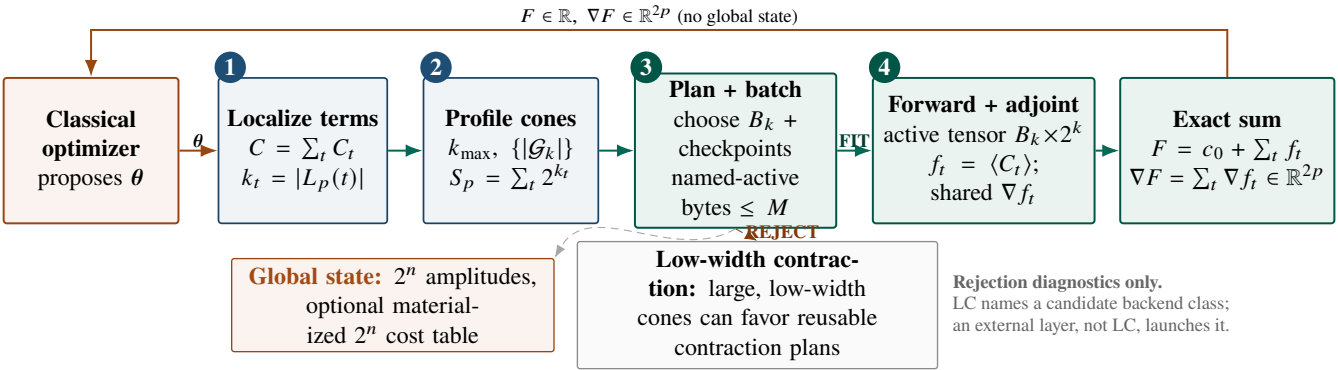
$F \in \mathbb{R}$, $\nabla F \in \mathbb{R}^{2p}$ (no global state)
Classical optimizer proposes θ
θ
1
Localize terms
$C = \sum_t C_t$
$k_t = |L_p(t)|$
2
Profile cones
$k_{\max}$, $\{|\mathcal{G}_k|\}$
$S_p = \sum_t 2^{k_t}$
3
Plan + batch
choose B_k + checkpoints
named-active bytes $\leq M$
FIT
4
Forward + adjoint
active tensor $B_k \times 2^k$
$f_t = \langle C_t \rangle$;
shared ∇f_t
Exact sum
$F = c_0 + \sum_t f_t$
$\nabla F = \sum_t \nabla f_t \in \mathbb{R}^{2p}$
REJECT
Global state: 2^n amplitudes, optional materialized 2^n cost table
Low-width contraction: large, low-width cones can favor reusable contraction plans
Rejection diagnostics only.
LC names a candidate backend class; an external layer, not LC, launches it.